

Matt Serdukoff

Phone: 978-239-6075 | Email: m.serdukoff@gmail.com
Website: mattserdukoff.co | GitHub: github.com/mserdukoff

PROFESSIONAL / TECHNICAL EXPERIENCE

TRAKUS:

Wakefield, MA

System Operator (*February 2021 - November 2022*)

- Held the position of System Operator at Trakus, a dynamic role that ensures the seamless operation of the tracking system during high-pressure equestrian racing events. Primary focus was on maintaining the system's functionality and performing data acquisition even under tight schedules, guaranteeing that it would consistently deliver top quality data. By closely monitoring and troubleshooting, I contributed to the precision and reliability of said data acquisition. The experience here honed my ability to thrive in fast-paced environments while maintaining a commitment to delivering exceptional results.

SELECTED PROJECTS:

Hime Skin Analysis Application

- Created a machine learning algorithm using a convolutional neural network to classify the images as either benign or malignant. Took part in project management and planning, AI enhanced system design, MLOps, data exploration, data analysis, and integrating the model with python web environments and iOS CoreML format.

Predicting Cardiovascular Disease using Machine Learning

- Using a dataset containing various symptoms and vitals, I wrote code in Jupyter Notebook that utilized Naïve Bayes, Logistic Regression, Linear Regression, and Decision Tree models to predict the presence of cardiovascular disease.

Learning a Children's Book using a Deep Averaging Network and Logistic Regression

- This project is from my Natural Language Processing class, it reads from a dataset that contains a diverse collection of text data from various children's books providing a valuable resource for NLP tasks tailored to young readers. Questions are categorized into four types: V (verbs), P (pronouns), NE (named entities), and CN (common nouns). For this project, only CN was utilized. Two models were tested, a Deep Averaging Network (DAN) and Logistic Regression.

Person or Dog Image Classifier

- Taking inspiration from the MNIST digit image classifier, I decided to see how accurate a neural network could be on data that is not polished. Using mass downloaded images from searches of both 'person' and 'dog', I created a model that can achieved an accuracy of 0.9911 with 20 epochs.

SKILLS

- | | | |
|--------------------------------------|--|-------------------------------|
| • C/C++, SFML | • Python | • SQL |
| • Pandas, NumPy, Matplotlib, PyTorch | • Scikit-learn, Natural Language Toolkit | • Natural Language Processing |
| • Machine Learning | • MLFlow | • Deep Learning |

RELEVANT COURSEWORK

Natural Language Processing

Machine Learning

Artificial Intelligence

CERTIFICATES

- Neural Networks and Deep Learning by DeepLearning.AI
- Natural Language Processing by DeepLearning.AI
- [Foundations of Data Science](#) (Part of Google Advanced Data Analytics Certificate)

EDUCATION

UNIVERSITY OF MASSACHUSETTS LOWELL:

Bachelor of Science in Computer Science with Data Science concentration

Lowell, MA

May 2024

ST. JOHN'S PREPARATORY SCHOOL:

High School Diploma

Danvers, MA

May 2019

LANGUAGES

Fluent: English and Russian

Proficient: Italian & German

Basic: Turkish and Hebrew